

The Complete Power BI Visualization Guide



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Stacked Bar Chart

Definitions

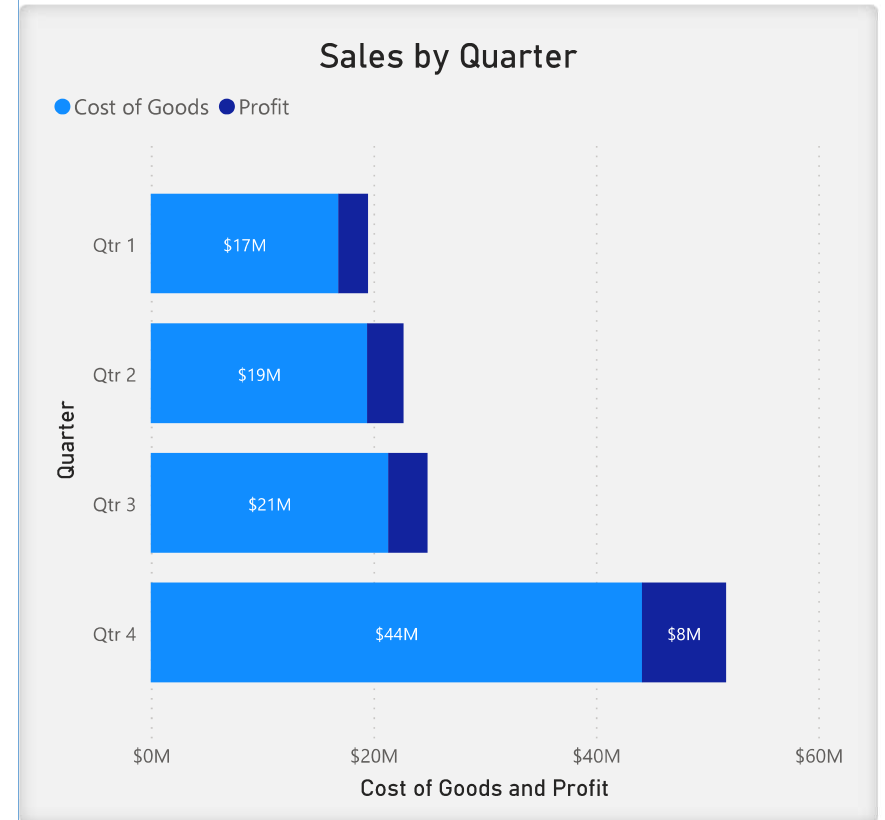
Similar to a Bar Chart, The Stacked Bar Chart displays numerical values over time or to compare values between different groups represented through rectangular bars on a graph. The Stacked Bar Chart provides a third dimension by also having categorical values within the bars themselves, separated by lines. Each bar in the chart represents a whole, and is divided into sub-bars that represent the different categories. This visual has the x-axis represented on the y axis and the y axis represented on x-axis. Use this visual when you want to compare numeric and categorical values together.

Use Cases

The Stacked Bar Chart can be used to display the Total Sales of a company by quarter. We can then further separate the bars into two categories displaying the proportion of each category to profit. In this visual we want to know how much of the Total Sales for each Category is Cost of Goods versus Profit.

Comparison

Part-to-whole



Stacked Column Chart

Definitions

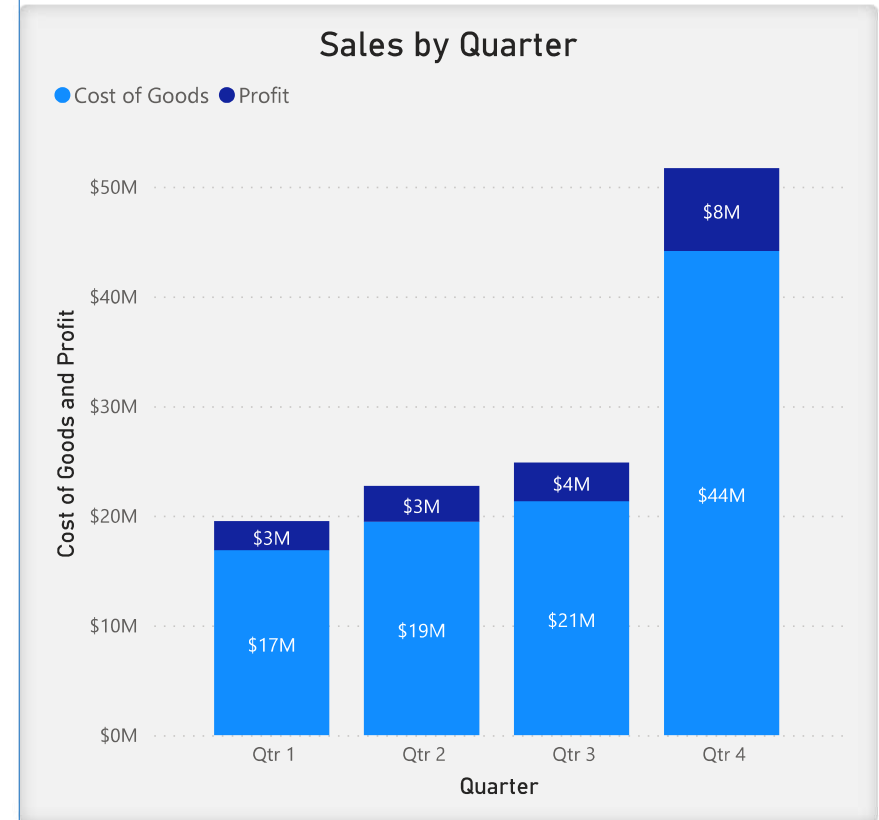
Similar to a Column chart, The Stacked Column Chart displays numerical values over time or to compare values between different groups represented through rectangular bars on a graph. The Stacked Column Chart provides a third dimension by also having categorical values within the bars themselves, separated by lines. Each bar in the chart represents a whole, and is divided into sub-bars that represent the different categories. Use this visual when you want to compare numeric values and categorical values together.

Use Cases

The Stacked Bar Chart can be used to display the Total Sales of a company by quarter. We can then further separate the bars into two categories displaying the proportion of each category to profit. In this visual we want to know how much of the Total Sales for each Category is Cost of Goods versus Profit.

Comparison

Part-to-whole



Clustered Bar Chart

Definitions

Similar to a Bar chart, the Clustered Bar Chart displays numerical values over time or to compare values between different groups represented through rectangular bars on a graph. The Clustered Bar chart also has several bars clustered next to each other representing another data series of categories in order to further compare the numerical versus categorical values.

Use Cases

The Clustered Bar Chart can be used to display the Total Sales of a company by quarter. We can represent Total Sales as two clustered bars where one represents the proportion of cost of goods to sales and the other represents the proportion of profit to sales.



Clustered Column Chart

Definitions

Similar to a Bar chart, the Clustered Column Chart displays numerical values over time or to compare values between different groups represented through rectangular bars on a graph. The Clustered Column chart also has several bars clustered next to each other representing another data series of categories in order to further compare the numerical versus categorical values.

Use Cases

The Clustered Column Chart can be used to display the Total Sales of a company by quarter. We can represent Total Sales as two clustered bars where one represents the proportion of cost of goods to sales and the other represents the proportion of profit to sales.



100% Stacked Bar Chart

Definitions

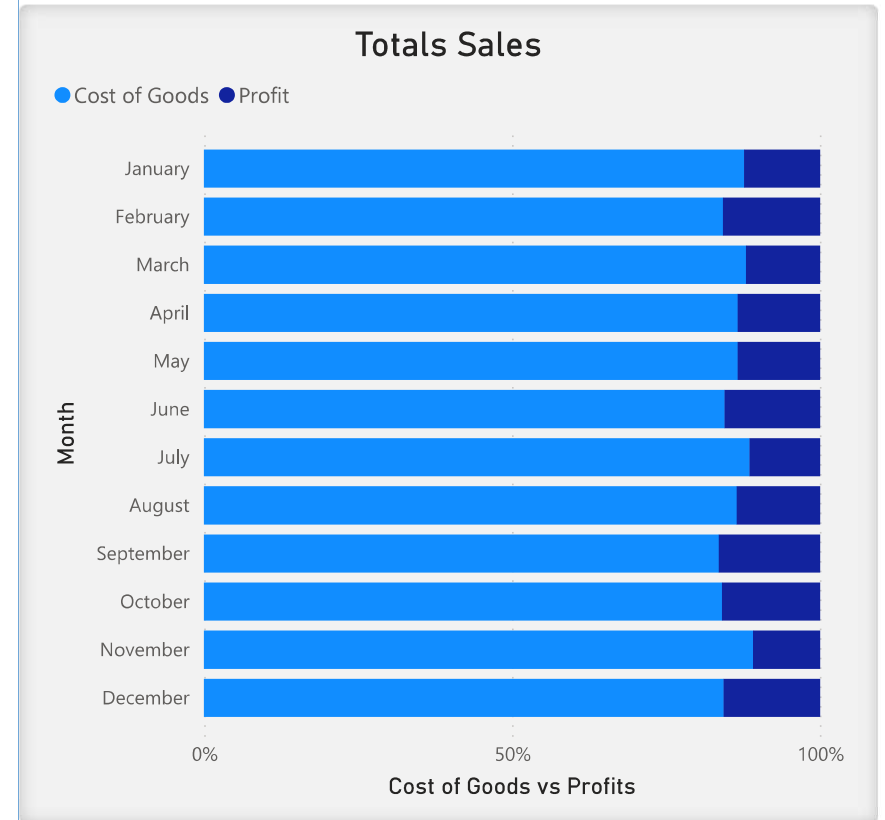
Similar to a Bar chart, the 100% Stacked Bar Chart displays numerical values over time or to compare values between different groups represented through rectangular bars on a graph. The 100% Stacked Bar Chart also displays percentages of the data as stacked bars where the total of all bars stacked together equals 100%.

Use Cases

The 100% Stacked Bar Chart can be used to display the Total Sales of company by quarter. Each bar represents a percentage of 100% of Total Sales. The bars are divided into two stacked bars where each stacked bar represents the percentage of Cost of Goods versus profit.

Comparison

Part-to-whole



100% Stacked Column Chart

Definitions

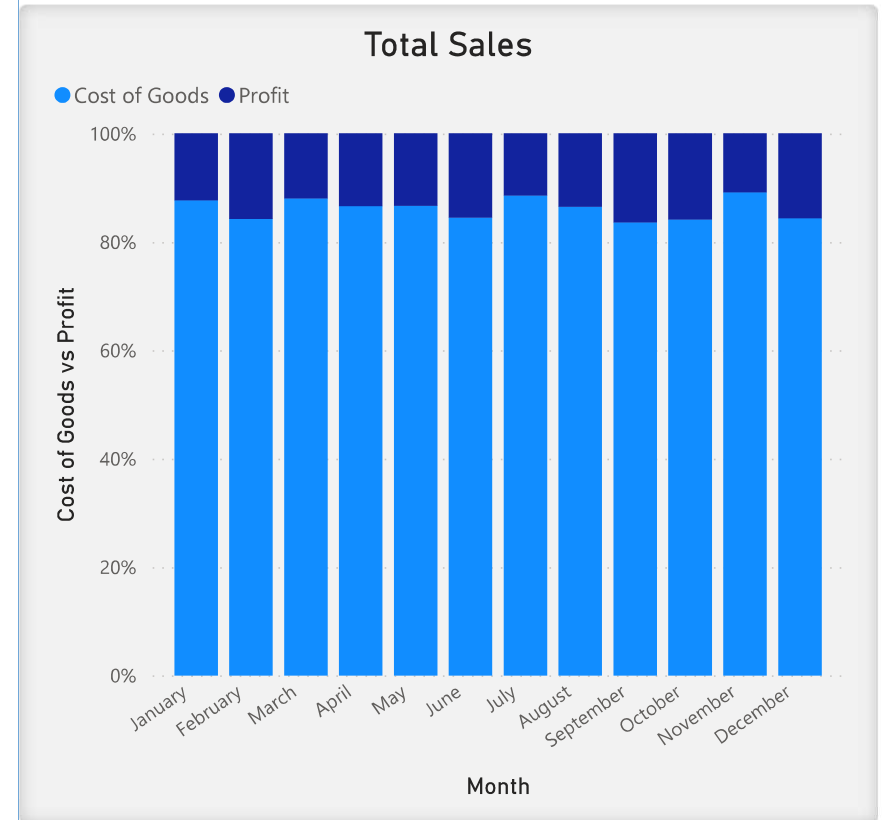
Similar to a Column chart, the 100% Stacked Column Chart displays numerical values over time or to compare values between different groups represented through rectangular bars on a graph. The 100% Stacked Column Chart also displays percentages of the data as stacked bars where the total of all bars stacked together equals 100%.

Use Cases

The 100% Stacked Column Chart can be used to display the Total Sales of company by quarter. Each bar represents a percentage of 100% of Total Sales. The bars are divided into two stacked bars where each stacked bar represents the percentage of Cost of Goods versus profit.

Comparison

Part-to-whole



Line Chart

Definitions

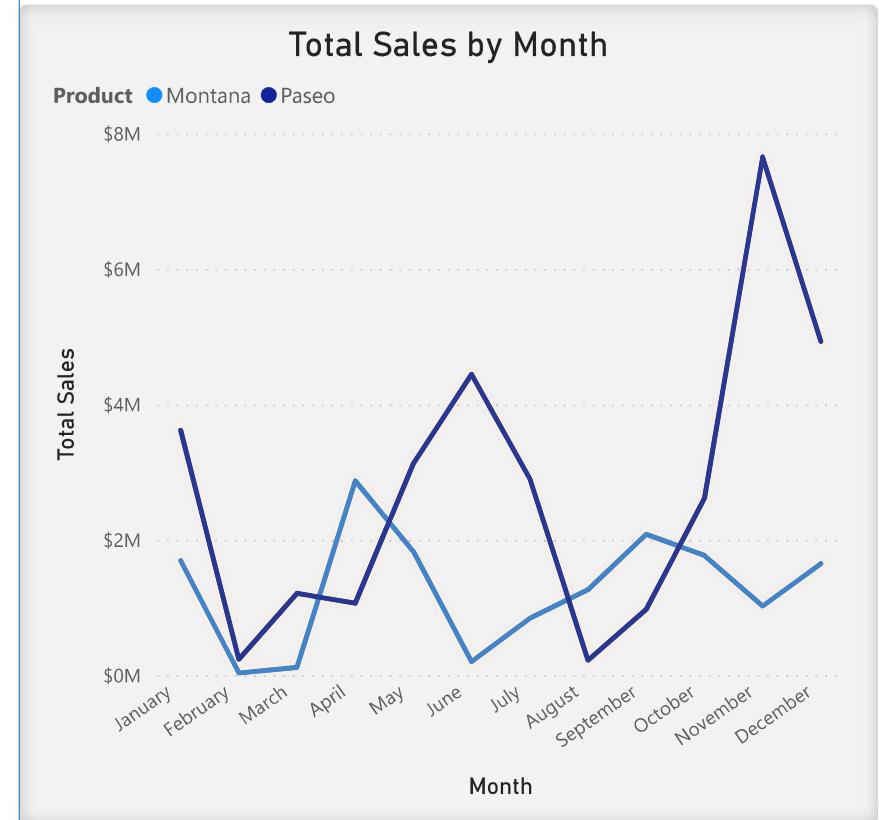
The Line Chart is a series of data points represented by dots and connected by straight lines on a graph. Line charts are used when you want to analyze progression over time. The Line Chart can have multiple data series displayed, represented as different lines and usually different colors.

Use Cases

The Line Chart can be used to analyze the trend of Profit and Cost of goods for a company on a monthly basis.

Distribution

Change over time



Area Chart

Definitions

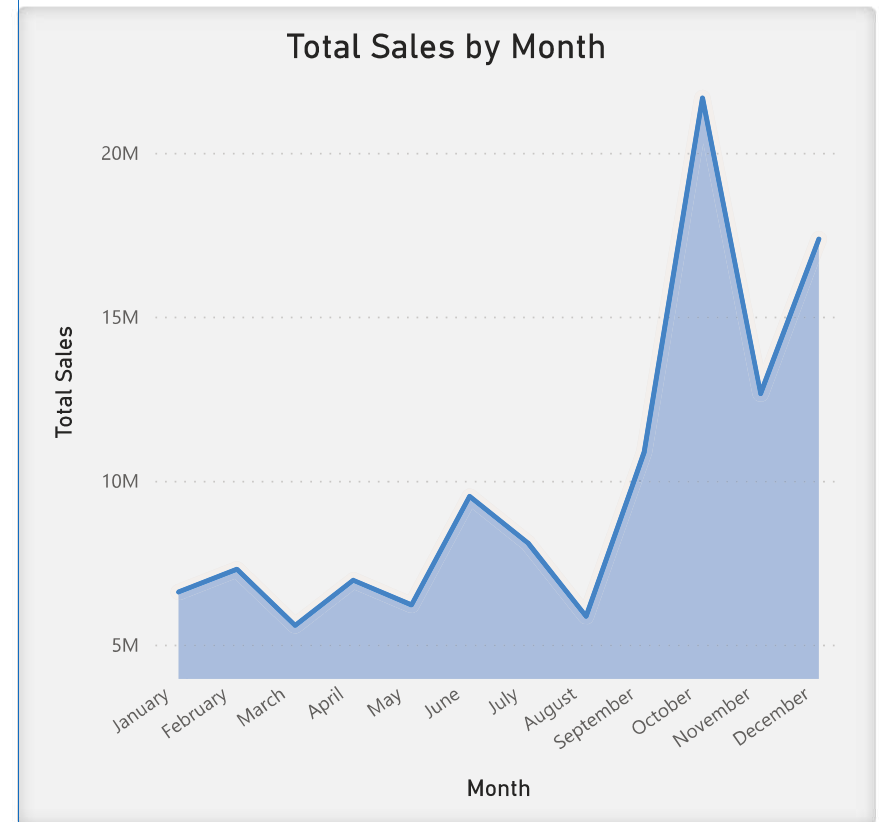
Similar to the Line Chart, the Area Chart is a series of data points represented by dots and connected by straight lines. The Area Chart also has the area between the axis and the line filled with color to demonstrate volume. Use this visual to if you want to display the change over time and also emphasise the magnitude of change with color.

Use Cases

The Area Chart can be used to analyze the trend of Total Sales for a company on a monthly basis. The Area Chart is best used for illustrating large differences between your values. The shaded area of The Area Chart emphasizes the changes of Total Sales month by month.

Comparison

Change over time



Stacked Area Chart

Definitions

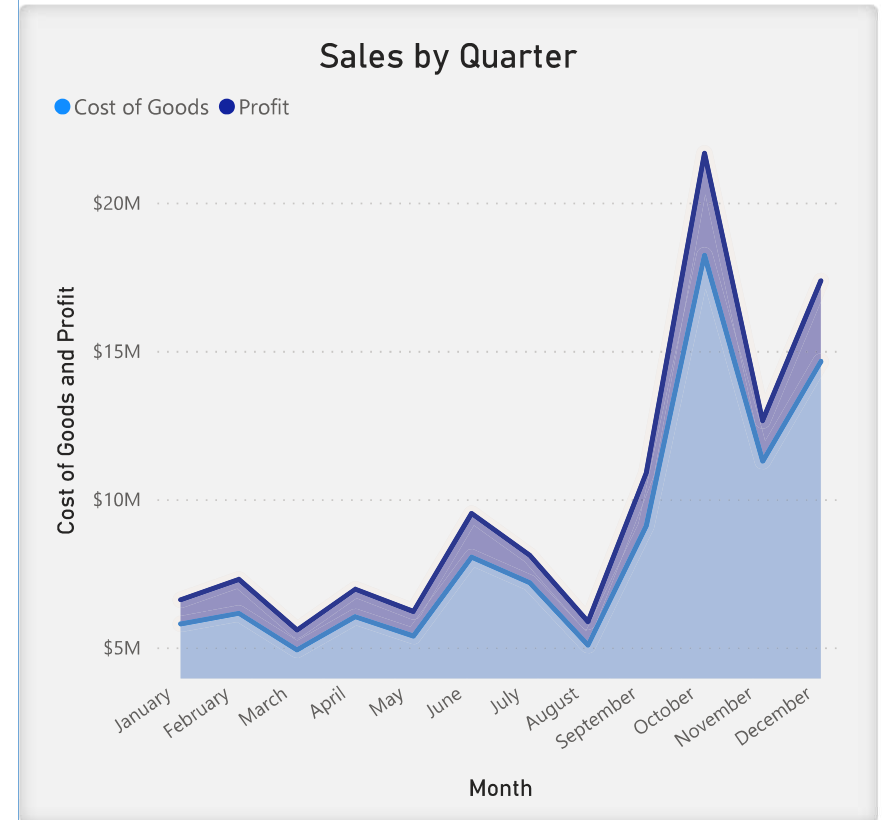
Similar to the Area Chart, the Stacked Area Chart is a series of data points represented by dots, connected by straight lines. The Area Chart has the area between the line and the axis shaded with color, and the shaded area is divided into stacked categories with each stacked area representing a proportional value of the Total.

Use Cases

The Stacked Area Chart can be used to analyze the Total Sales over for a company on a monthly basis. The area of the chart can be divided into stacked areas representing the proportion of Cost of goods and Profit to the Total Sales of the company.

Change over time

Part-to-whole



Line and Stacked Column Chart

Correlation

Change over time

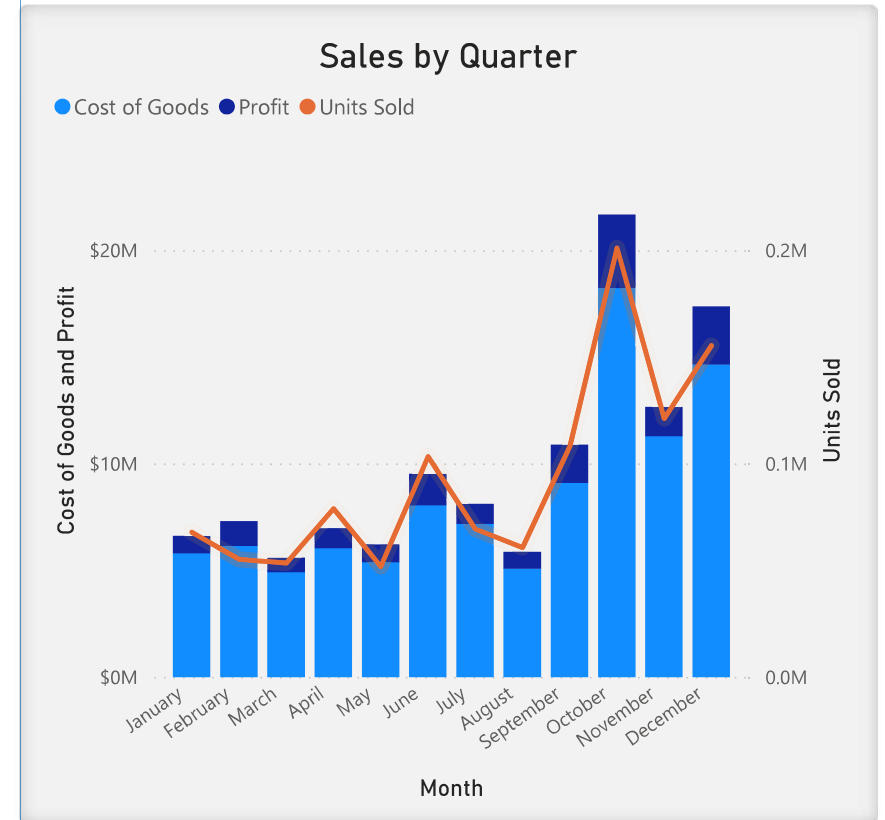
Part-to-whole

Definitions

The Line and Stacked Column Chart combines a Line Chart and a Stacked Column Chart together into one visual. This allows the user to make a quicker comparison of two different categories. A Line chart is a series of data points represented by dots and connected by straight lines that is used to progression over time. The Stacked Column Chart displays numerical values over time or to compare values between different groups represented through rectangular bars on an graph. The Stacked Column chart also displays the bars divided into stacked bars that display the proportional values of a another dimension.

Use Cases

The Line and Stacked Column Chart can be used to compare Total Sales and the Number of Units sold. The Total Sales over a monthly basis will be your Stacked Column Chart, with the stacked bars divided into Total Profit and Cost of goods in proportion to Total Sales. The Line Chart on top of the Clustered column chart, can represent the progression of Total profit over the same time period.



Line and Clustered Column Chart

Correlation

Comparison

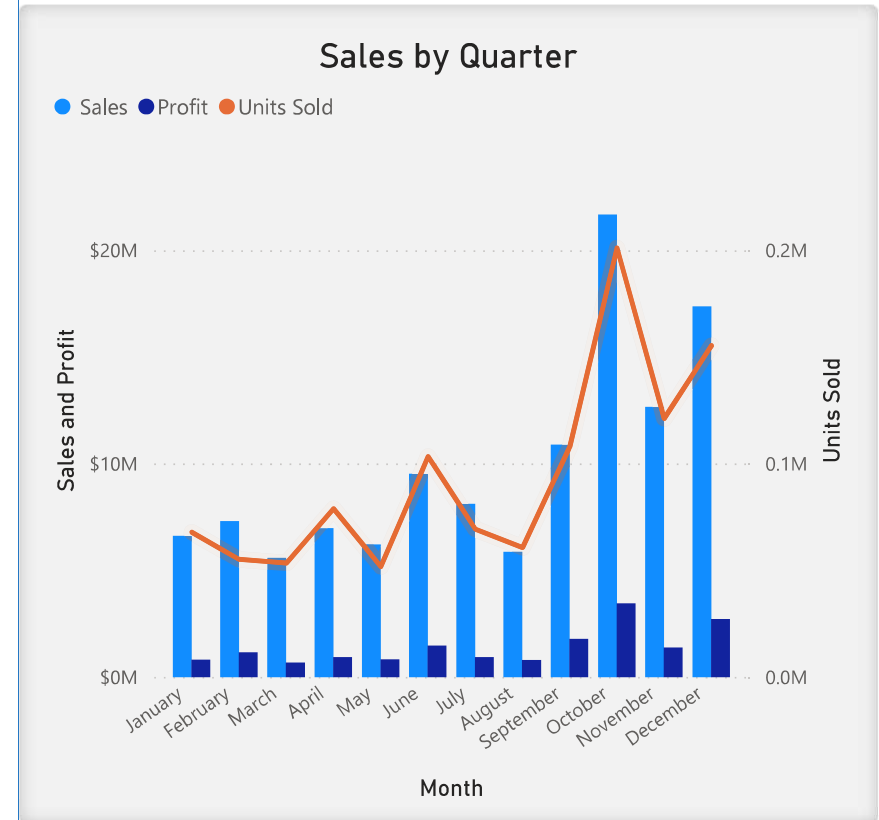
Part-to-whole

Definitions

The Line and Clustered Column Chart combines the Line Chart and Clustered Column Chart into one visual. A Line chart is a series of data points represented by dots and connected by straight lines that displays trends over time. Similiar to a Bar chart, a Clustered Column Chart displays numerical values over time or to compare values between groups, represented through rectangular bars on a graph. The Clustered bar chart also has several bars next to each other representing another series of data categories in order to compare the numerical and categoricial values. Use the Line and Clustered Column Chart when you want to make a quick comparison between two data series.

Use Cases

The Line and Clustered Column Chart can be used to compare Total Sales and the Number of Units sold. Total Sales over a monthly basis will be your Clustered Column Chart, with two clustered bars representing the proportions of cost of goods and profit to Total Sales. The Line Chart on top of the Clustered column chart, can represent the progression of Total profit over the same time period.



Ribbon Chart

Definitions

The Ribbon Chart displays ranked data categories illustrated through ribbons, where the highest rank is always displayed on the top ribbon for each time period. This Ribbon chart helps visualize how a given category ranks throughout a time period.

Use Cases

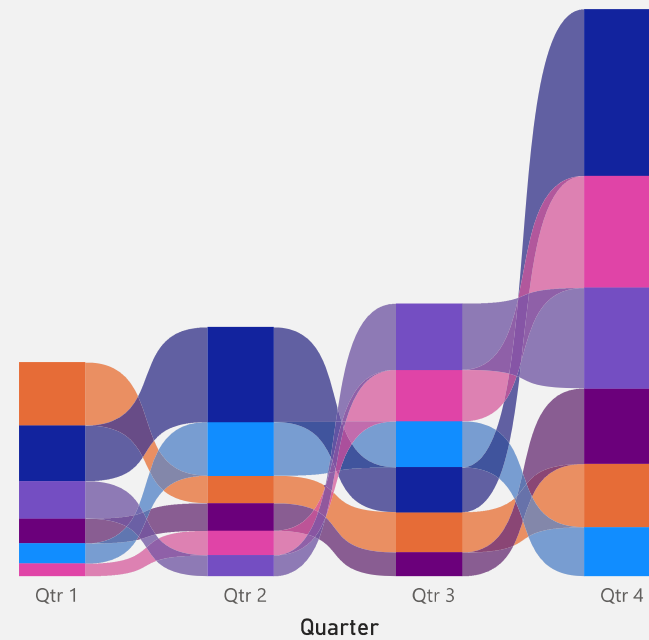
Use the Ribbon Chart for an example such as ranking Total Sales for Product sold on a Quarterly basis. The Ribbon chart can visualize what company product is ranked as selling the most for each quarter.

Comparison

Part-to-whole

Total Sales by Quarter

Product ● Amarilla ● Carretera ● Montana ● Paseo ● Velo ● VTT



Waterfall Chart

Definitions

The Waterfall Chart illustrates how an initial value is affected by positive and negative changes. The chart shows a running total as values are added or subtracted. The values are shown as bars on a graph with the direction of the bars representing negative or positive values. The bars are also color coded, which allows the user to quickly notice the increases and decreases in their data.

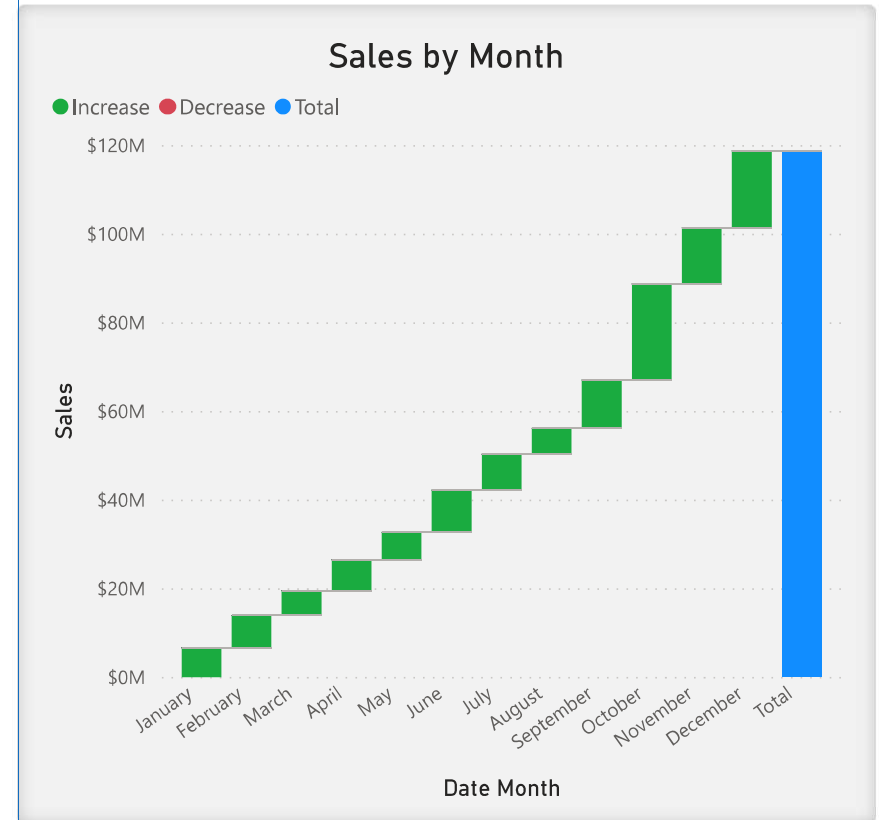
Use Cases

The Waterfall Chart can be used to illustrate changes in Total Sales for a company on a monthly basis. The company can use the chart to visualize which months do the Total Sales increase or decrease.

Flow

Change over time

Part-to-whole



Funnel Chart

Definitions

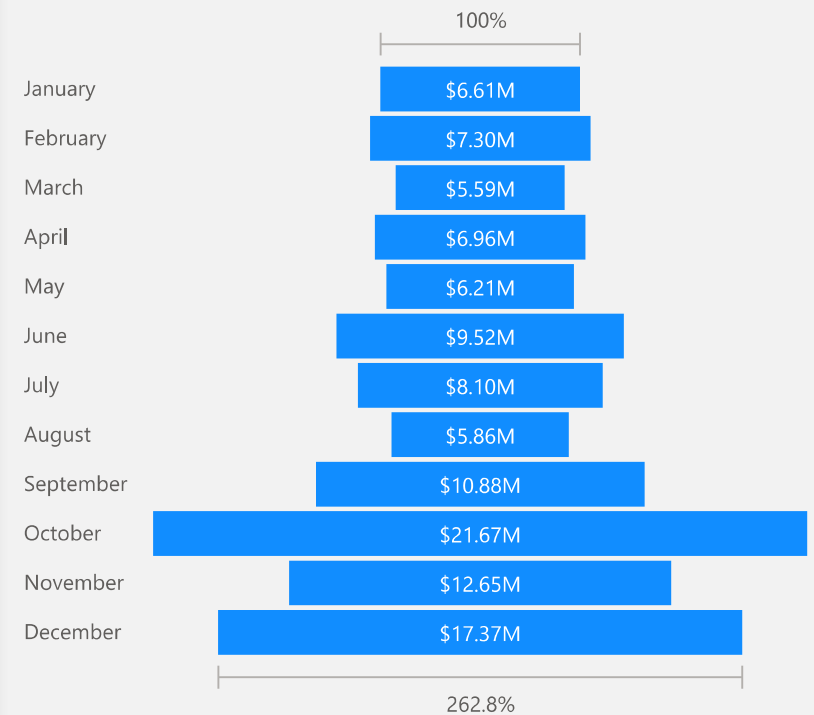
This Funnel Chart displays a linear process that has sequential connected stages. The chart communicates to the user the health of the process you are tracking. Each stage of the funnel chart, illustrated by a rectangle, represents a percentage of the total. Use the funnel chart when your data is sequential through at least 4 stages and also if the first stage is expected to be a greater percentage than the final stage. This visual also helps in determining bottlenecks in your workflow.

Use Cases

Use the Funnel Chart for an example such as tracking customer leads. The Funnel chart can visualize the sequential process of how many leads, prospects or contracts of customers a company has. The number of customers within each process will be represented by the size of the rectangle.

Flow

Sales by Month



Scatter Chart

Correlation

Definitions

The Scatter Chart displays the relationship between two numerical values. The Scatter Chart combines the intersections of x and y into single data points. This chart will display the correlation between two numerical values. If you want to add a third dimension, you can replace the data points with bubbles(circles), with the size of each bubble represented by the third dimension.

Use Cases

The Scatter Chart can be used to display the relationship between Profit and Discounts for each product sold at a company. This visual can illustrate the affect of implementing a discount on a product and further analyzing whether the discount affects Total Profit.



Pie Chart

Part-to-whole

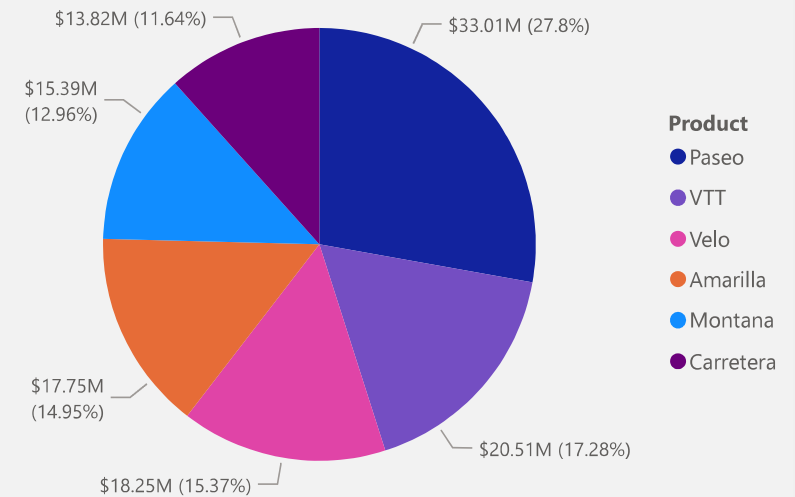
Definitions

The Pie Chart displays the relationship of parts to whole by illustrating numerical proportions represented as slices of a pie where the entire pie represents the whole. This Chart displays the relationship of parts to a whole.

Use Cases

The Pie Chart can be used to display the proportion of Total Sales related to each product. The Largest portion of the pie is the product that produces the most sales for the company.

Sales by Product



Donut Chart

Part-to-whole

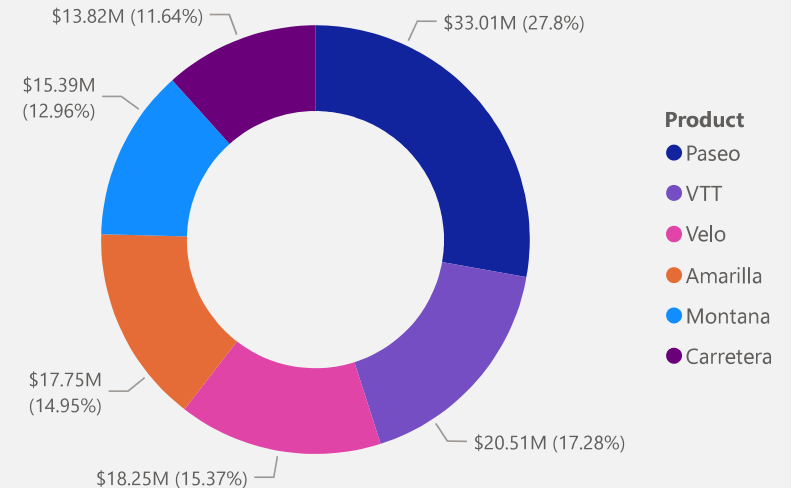
Definitions

Similar to the Pie Chart, the Donut Chart illustrates numerical proportions represented as slices of a pie where the entire pie represents the whole. This chart displays the relationship of parts to a whole. Unlike the Pie Chart, the Donut Chart has a circular center that is cut out of the pie, allowing blank space for a label or icon.

Use Cases

The Donut Chart can be used to display the proportion of Total Sales related to each product. The Largest portion of the pie is the product that produces the most sales for the company.

Sales by Product



Treemap

Part-to-whole

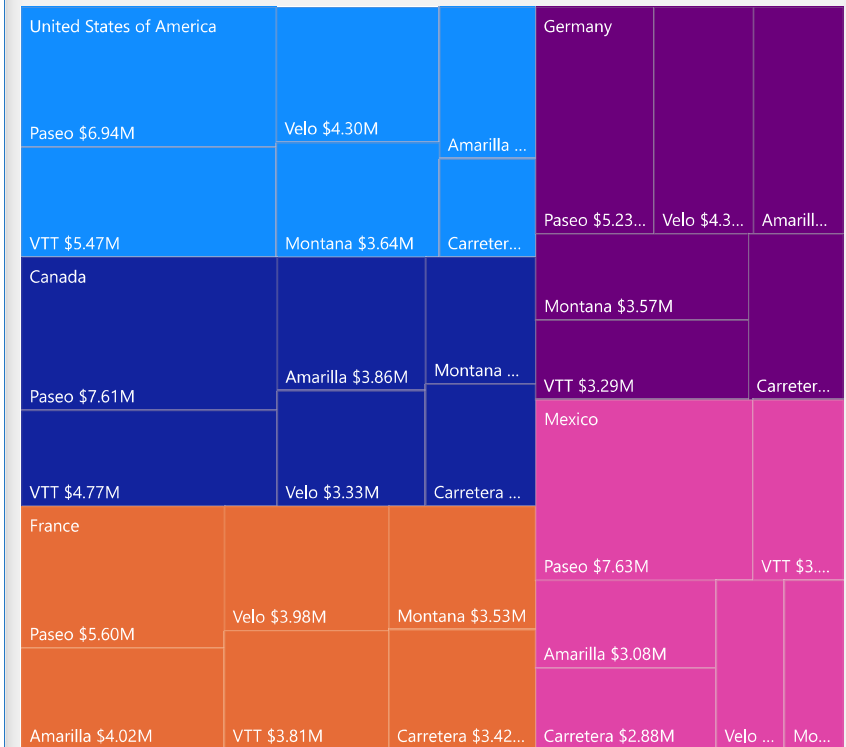
Definitions

The Treemap visual displays hierarchical data as a set of nested rectangles. The levels of the hierarchy are represented by a colored rectangle also known as the branch. The rectangles within the rectangles further categorize the data and are known as the leaves. The size of each rectangle is determined by a metric and are ordered from largest to smallest. Use a Treemap if you have a lot of hierarchical data to display and to show proportional relationships between each part to a whole.

Use Cases

The Treemap visual can be used to display the proportion of Total sales related to each product. The rectangles can represent products and the size of the rectangle can be represented by the proportion of that product to the Total sales of the company.

Sales by Country and Product



Map

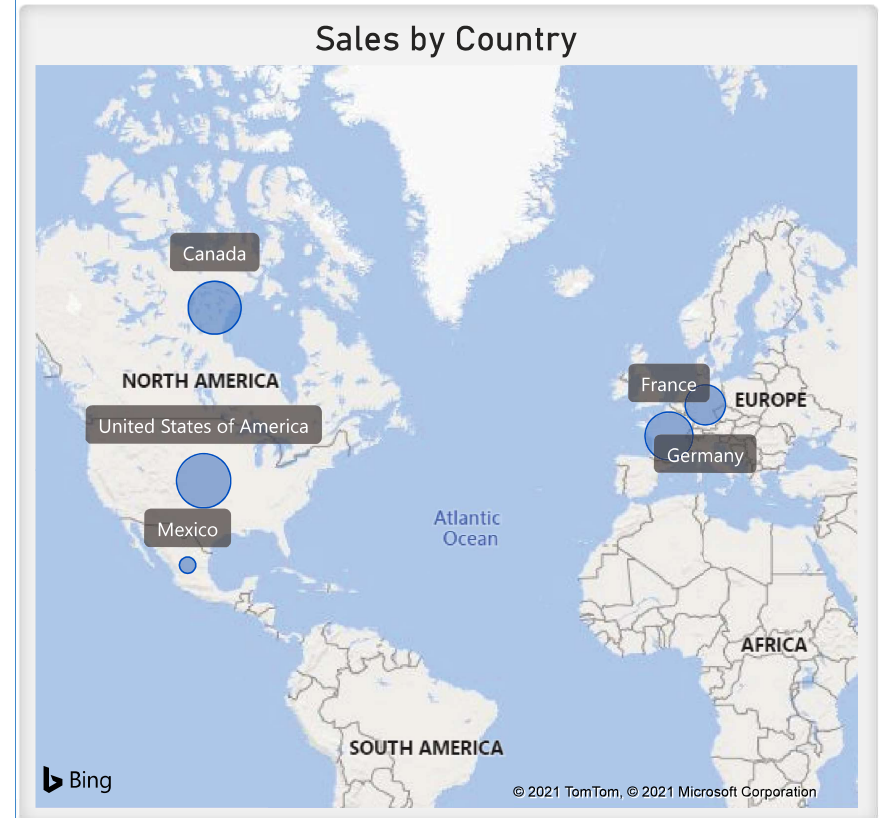
Spatial

Definitions

The Map visual maps out data points that have longitude and latitude coordinates as their values. You can include a third dimension to this visual by representing the data points using bubbles(Circles), where the size of each bubble is represented by the value of your third metric.

Use Cases

The Map visual can be used to visualize which countries a company makes sales in. The size of the data point could represent the amount of sales from that country. This visual quickly indicates which country has the most sales.



Filled Map

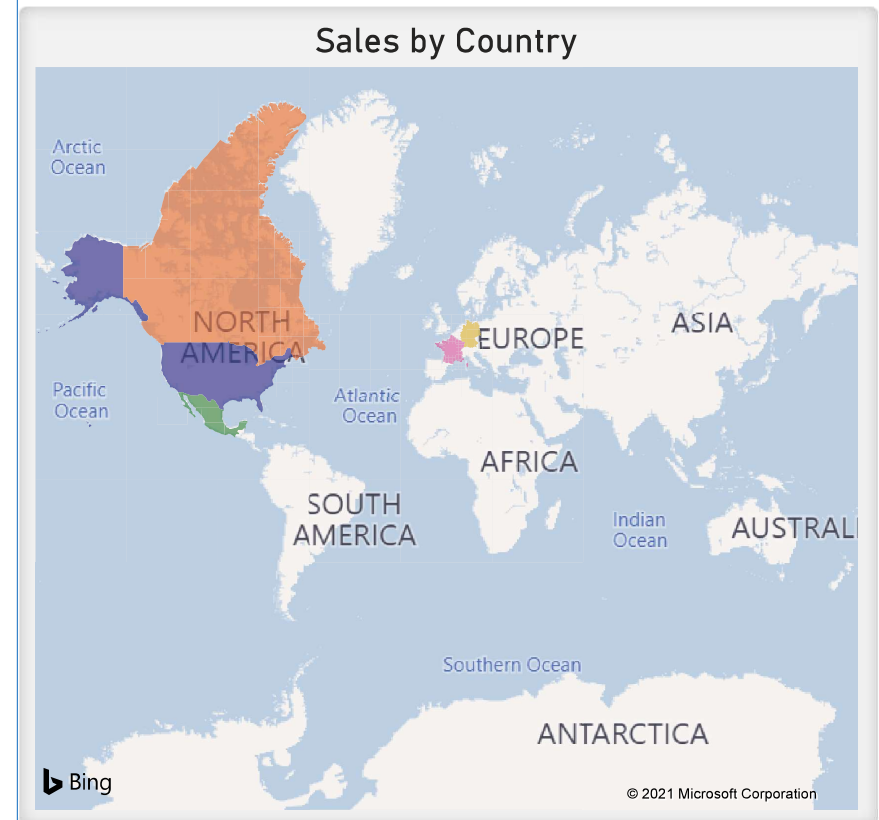
Definitions

The Filled Map visual displays how a value differs in proportion across a geography or region by the use of shading or tinting or patterns.

Use Cases

The Filled Map visual can be used to display the countries a company makes sales in. The geographical regions that produce sales can be shaded in different colors.

Spatial



Definitions

Similar to the Map visual, The ArcGIS map displays data points on a map referencing values that have longitude and latitude coordinates. The ArcGIS map also has a variety of themes, reference layers, symbol styles, location types and base maps to choose from. Use the ArcGIS map to create maps of deeper understanding.

Use Cases

Use the ArcGIS map for examples such as identifying location in relation to other critical variables such as population size.





Comparison

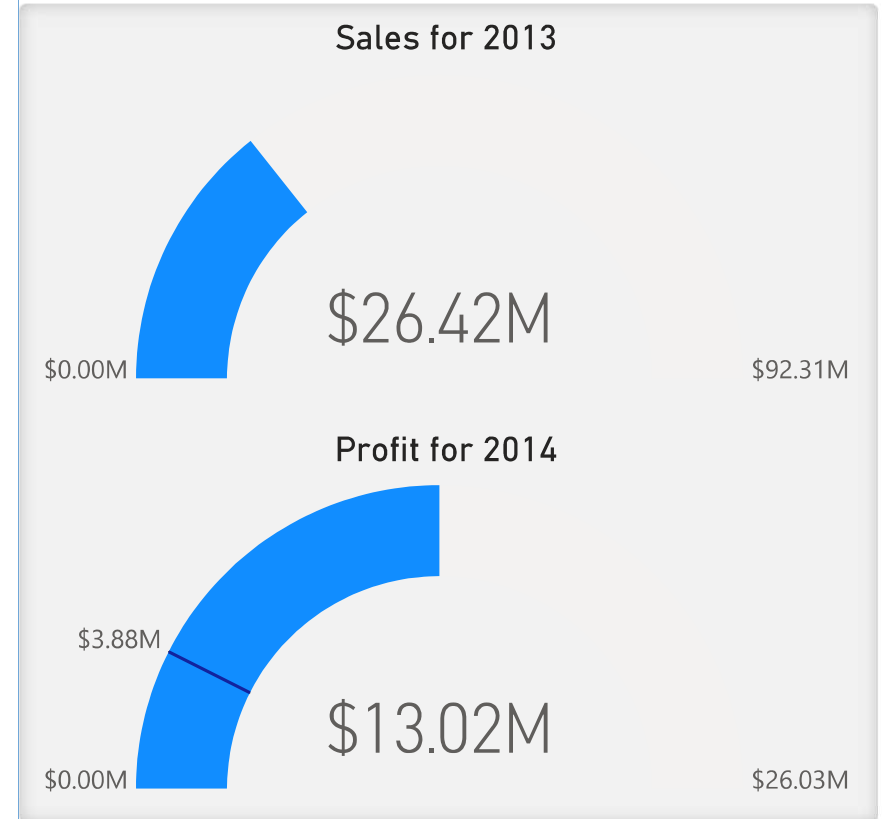
Single

Definitions

The Gauge visual displays the minimum, maximum and current value against a goal or Key Performance Indicator. The line(or needle) on the gauge visual represents the goal or KPI. The shading of the gauge and the value inside the arc represent the progress value. Use the gauge visual when you want to measure progress towards a particular goal.

Use Cases

The Gauge visual can be used to display Total Sales currently, a target sales goal and the Total Sales from last year. The visual illustrates how far current Total Sales is from a target sales goal.



Card

Single

Definitions

The Card visual is a single value displayed in a rectangular box. Use the card visual when you have a critical value that is important to communicate to your audience.

Use Cases

Use the Card visual for displaying important values for business decisions such as Total Sales, Total Sales for the year before or Profit.

\$118.73M

Total Sales



Multi-Row Card

Ranking

Single

Definitions

Simliar to the Card Visual, The Multi-Row Card visual is used to display single values but in a group format or in a section. Use this visual if you have multiple single values that you want to present in your report.

Use Cases

The Multi-Row Card can be used as to display a series values showing Sales and Profit and grouping them by Product.

Paseo

\$33,011,143.95

Sales

\$4,797,437.95

Profit

VTT

\$20,511,921.02

Sales

\$3,034,608.02

Profit

Velo

\$18,250,059.47

Sales

\$2,305,992.47

Profit

Amarilla

\$17,747,116.06

Sales

\$2,814,104.06

Profit

Montana

\$15,390,801.88

Sales

\$2,114,754.88

Profit

Carretera

\$13,815,307.89

Sales

\$1,826,804.89

Profit





Change over time

Single

Definitions

The KPI visual is a visual cue that communicates the progress made towards a measurable goal. It contains the goal and the current progress as single values. You can also add a trend axis to the visual that displays an area chart behind the values to represent the trend of progress over time.

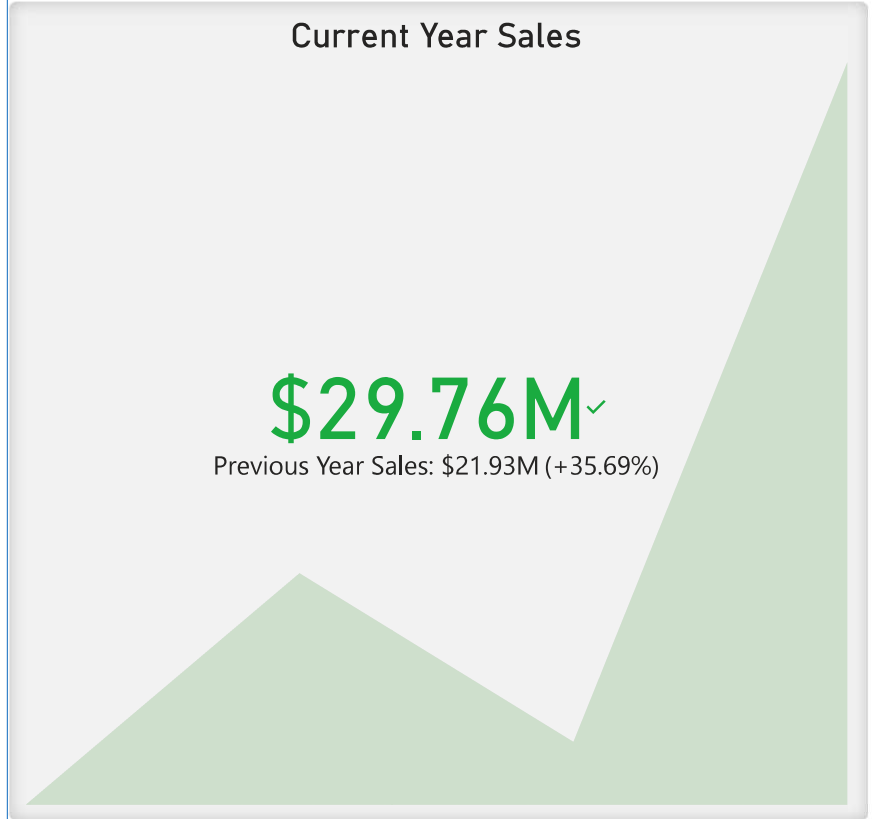
Use Cases

The KPI visual can be used to measure progress towards a sales goal. The KPI visual can display Total Sales currently, Total Sales of the previous year and a trend line to show the progression over time.

Current Year Sales

\$29.76M✓

Previous Year Sales: \$21.93M (+35.69%)





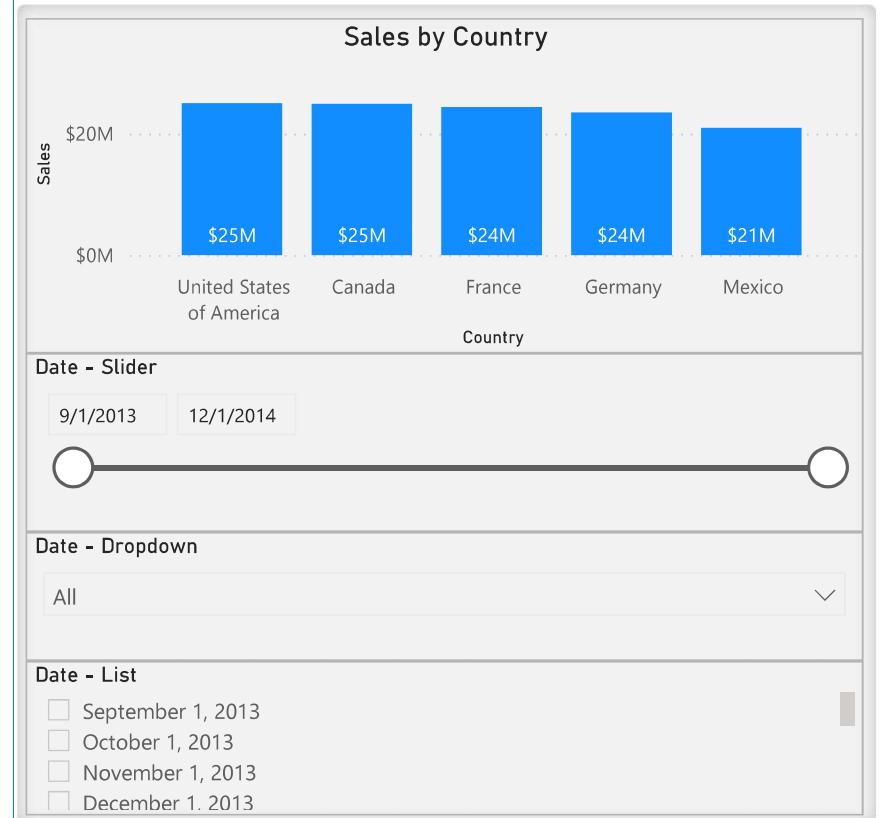
Definitions

Similar to the filtering pane, the Slicer can filter your data. You can select how you want to filter the dataset directly from the report canvas. The slicer will affect the other visuals on the report based on the chosen filter. Use the slicer to create more focused reports by displaying common or important filters.

Use Cases

The Slicer can be used to display a Country Name slicer that drills down the visuals in the report to the selected country or countries.

Other





Definitions

This basic 2-dimensional table represents tabular data in a grid-like format. It is a flat structure representation of the data.

Use Cases

The Table visual can be used to display a more detailed view of transactions within a company.

Comparison

Ranking

Single

Product	Year	Quarter	Month	Country	Sales	Profit
Amarilla	2014	Qtr 1	January	Canada	\$180,416.25	(\$18,663.75)
Amarilla	2014	Qtr 1	February	Canada	\$705,600.00	\$123,200.00
Amarilla	2014	Qtr 1	March	Canada	\$229,104.00	\$7,104.00
Amarilla	2014	Qtr 2	April	Canada	\$38,021.40	\$27,459.90
Amarilla	2014	Qtr 2	May	Canada	\$191,231.25	(\$6,168.75)
Amarilla	2014	Qtr 2	June	Canada	\$20,416.05	\$7,661.05
Amarilla	2014	Qtr 2	July	Canada	\$22,256.33	\$5,951.33
Amarilla	2014	Qtr 3	August	Canada	\$183,540.00	\$40,020.00
Amarilla	2014	Qtr 3	September	Canada	\$222,705.00	\$38,885.00
Amarilla	2014	Qtr 4	October	Canada	\$421,017.94	\$77,407.94
Amarilla	2014	Qtr 4	November	Canada	\$20,794.80	\$9,614.80
Amarilla	2014	Qtr 4	December	Canada	\$563,304.00	\$84,304.00
Carretera	2014	Qtr 1	January	Canada	\$32,370.00	\$16,185.00
Carretera	2014	Qtr 1	February	Canada	\$20,687.16	\$14,876.16
Carretera	2014	Qtr 1	March	Canada	\$281,053.50	\$41,073.50
Carretera	2014	Qtr 2	April	Canada	\$90,956.25	\$1,856.25
Carretera	2014	Qtr 2	May	Canada	\$16,121.40	\$7,811.40
Carretera	2014	Qtr 2	June	Canada	\$338,762.40	\$36,822.40
Carretera	2014	Qtr 3	July	Canada	\$5,217.03	\$1,122.03
Carretera	2014	Qtr 3	August	Canada	\$21,025.44	\$15,373.44
Carretera	2014	Qtr 3	September	Canada	\$17,166.60	\$12,831.60
Carretera	2014	Qtr 4	October	Canada	\$401,197.20	\$23,312.20
Carretera	2014	Qtr 4	November	Canada	\$35,494.80	\$8,604.80
Carretera	2014	Qtr 4	December	Canada	\$1,007,006.00	\$212,206.00
Total					\$118,726,350.26	\$16,893,702.26



Definitions

Similar to a table, the Matrix represents tabular data in a grid-like format. It also automatically aggregates the data and enables you to drill down. The matrix displays data meaningfully across multiple dimensions. To gain insights quicker, you can also apply conditional formatting.

Use Cases

The Matrix visual can be used to display a more detailed view of Sales, Profit and Cost of goods for each product, grouped by Country.

Comparison

Ranking

Single

Country	Sales	Profit	Cost of Goods
United States of America	\$25,029,830.17	\$2,995,540.67	\$22,034,290
Paseo	\$6,944,325.77	\$1,020,603.27	\$5,923,723
VTT	\$5,465,592.64	\$647,896.64	\$4,817,696
Velo	\$4,298,828.00	\$265,401.00	\$4,033,427
Montana	\$3,644,126.80	\$434,521.80	\$3,209,605
Amarilla	\$2,837,117.41	\$388,626.41	\$2,448,491
Carretera	\$1,839,839.55	\$238,491.55	\$1,601,348
Canada	\$24,887,654.89	\$3,529,228.89	\$21,358,426
Paseo	\$7,611,520.99	\$1,265,017.99	\$6,346,503
VTT	\$4,768,754.31	\$488,808.81	\$4,279,946
Amarilla	\$3,855,765.88	\$646,861.38	\$3,208,905
Velo	\$3,329,490.34	\$370,568.34	\$2,958,922
Montana	\$2,711,919.03	\$321,867.03	\$2,390,052
Carretera	\$2,610,204.34	\$436,105.34	\$2,174,099
France	\$24,354,172.28	\$3,781,020.78	\$20,573,152
Paseo	\$5,597,751.06	\$838,748.56	\$4,759,003
Amarilla	\$4,016,427.13	\$667,867.63	\$3,348,560
Velo	\$3,978,096.24	\$707,930.24	\$3,270,166
VTT	\$3,811,193.59	\$716,371.09	\$3,094,823
Montana	\$3,527,382.37	\$461,238.37	\$3,066,144
Carretera	\$3,423,321.90	\$388,864.90	\$3,034,457
Germany	\$23,505,340.82	\$3,680,388.82	\$19,824,952
Total	\$118,726,350.26	\$16,893,702.26	\$101,832,648

R script visual

Other

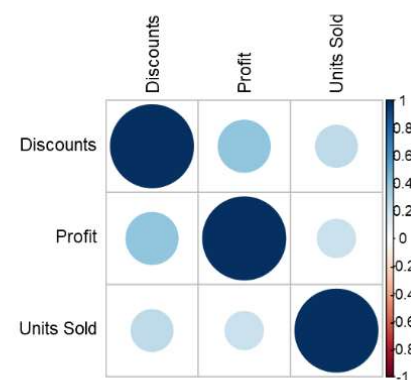
Definitions

The R visual uses an R script. It allows you to prepare, analyze, and plot visuals by running R scripts on your dataset directly on the Power BI desktop. However, you can only run R scripts if you have downloaded R to your local computer.

Use Cases

Use R visuals when you want to go beyond the out of the box visuals and create visuals such as correlation plots, violin graphs, and heatmaps.

Correlation Plot



```
150 #PBI_COMMENT: Create visual
151 if(nc > 1 && nr > 1){
152   #PBI_COMMENT: compute correlation matrix, allow for missing values
153   M <- cor(dataset, use="pairwise.complete.obs")
154   corplot(M, method=method, order=order, type=type, addrect=addrect,
155           mar = defMar, tl.col = tl.col, tl.cex=tl.cex,
156           number.digits=number.digits, number.cex=number.cex, addCoef.col=addCoef.col)
157 }else{ #empty correlation plot
158   plot.new()
159   pbiWarning<-paste(pbiWarning, "Not enough input dimensions", sep="\n")
160 }
161 #add warning as subtitle
162 if(showWarnings)
163   title(main = NULL, sub = pbiWarning, outer = FALSE, col.sub = "gray60", cex.sub = 0.75)
```

Correlation Plot

Bar Chart



Python visual

Other

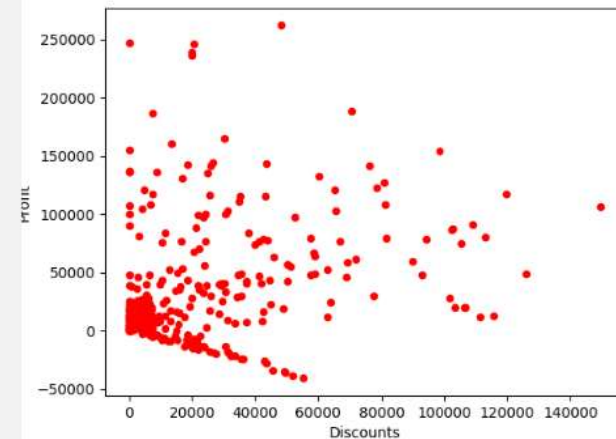
Definitions

The Python visual uses the Python programming language. It allows you to prepare, analyze, and plot visuals by running Python scripts on your dataset directly on the Power BI desktop. However, you can only run Python scripts if you have downloaded Python to your local computer.

Use Cases

Use a Python visual when you want to go beyond the out of the box visuals and create visuals such as pair plots, or circular bar graphs.

Scatterplot



```
7 import matplotlib.pyplot as plt
8 dataset.plot(kind='scatter', x='Discounts', y='Profit', color='red')
9 plt.show()
```

Scatterplot

Pie Chart

Bar Chart



Key Influencers

AI Visual

Definitions

The key Influencer visual analyzes the data and ranks in order of factors that matter. Therefore, the factors with the highest correlation are displayed first. Use this visual to understand what factors affect the metric being analyzed or when you want to contrast the relative importance of the factors.

Use Cases

The Key Influencer visual can be used to determine what influences Profit to increase for a company. The visual indicates that the more Sales increases the more Profit Increases and the more Discounts decreases the more Profit increases.

Key influencers Top segments



What influences Profit to Increase ?

When...

....the average of Profit increases by

Sales goes up 1464260.10

\$158.5K

Discounts goes down
145540.92

\$139.0K

Gross Sales goes up
1589917.72

\$120.9K

Units Sold goes up
14272.74

\$36.78K



Decomposition Tree

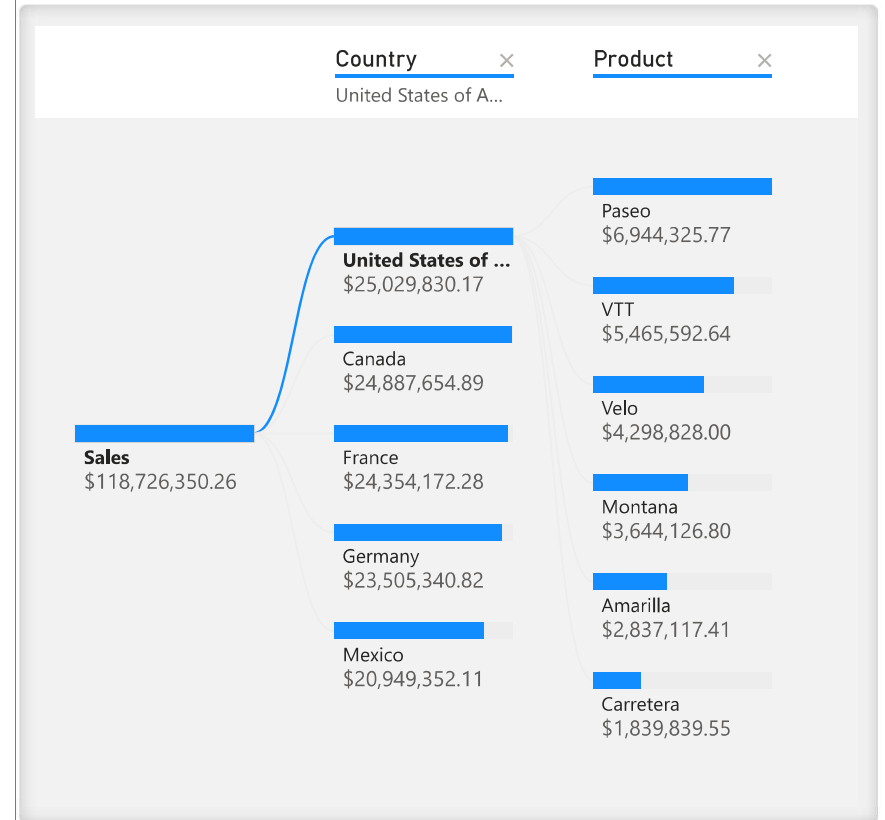
AI Visual

Definitions

The Decomposition Tree visual displays your data across multiple dimensions by aggregating the data for you, enabling you to drill down in any order. It uses artificial intelligence (AI) to find the next dimension to drill down. Use the Decomposition Tree when you want to conduct root cause analysis or ad hoc exploration.

Use Cases

The Decomposition Tree visual can be used to illustrate Sales being displayed across the dimensions of Country and Product.





AI Visual

Definitions

Q&A allows you to type a question in the report using natural language and receive an answer as a visual. The Q&A engine uses artificial intelligence (AI) to decide which visual to use based on the words in the question.

Use Cases

Use the Q&A feature to type questions in natural language such as "What is the top Country by Total Sales?" or "What Product has the most Sales?"



Try one of these to get started

top countries by total sales

what is the total sales by country

[Show all suggestions](#)



Custom Visuals

Other

Definitions

The "Get more visuals" feature takes you to a Power BI visual store where you can add custom visuals made by other developers or from your organization. You can add the custom visuals to the selection of visuals already in Power BI.

Use Cases

Use the "Get more visuals" feature when you want to add custom visuals such as Word Cloud, Tornado charts, Radar charts, or Road map to your dashboard.

